

WASP-12b

R-1425

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_161104 · created 2026-08-01T06:48:58+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457695.92314 +/- 0.00033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.045
Transit depth (Rp/Rs)^2	2.39 +/- 1.38 [%]
Orbital Inclination (inc)	80.31 +/- 2.86
Airmass coefficient 1 (a1)	1.12 +/- 0.54
Airmass coefficient 2 (a2)	-0.05 +/- 0.41
Scatter in the residuals of the lightcurve fit is	47.4 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.1002 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.155 $\pm$ 0.045 vs 0.1178 (deviation 0.83 σ)
Duration fit vs published	0.1002 d vs 0.125 d (deviation 4.2 σ)
Mid-time O-C	-1571.4 min
Depth SNR	3.4
Residual scatter	47.4 %

Light curve

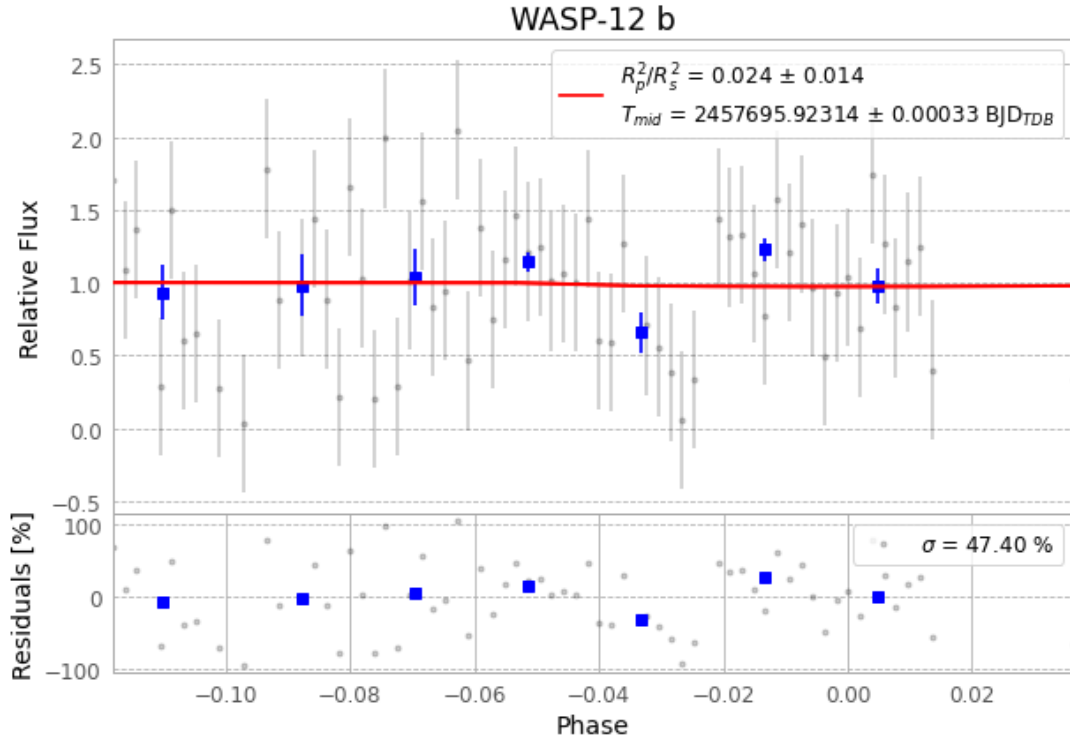


Figure 1 — FinalLightCurve_WASP-12 b_2016-11-02.png (SHA-256 dbb36c9b99c418eb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 436], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 90f4070f13683d2f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-12161103065712.FITS → WASP-12161103115712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161104.log (e5f6a8b7f624...), FinalLightCurve_WASP-12 b_2016-11-02.png (dbb36c9b99c4...), FinalLightCurve_WASP-12 b_2016-11-02.csv (da5ba466cc26...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 06:48Z.